Reducing Cognitive Load in Clinical Decision-Making with GenAI-Powered Report Analysis

— a prototype and study proposal

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INTRODUCTION

Clinicians managing complex patients must manually navigate multiple longitudinal test reports — a cognitively demanding process that makes it difficult to identify trends and synthesise findings in the point of care. The burden of integrating these findings can lead to missed insights and suboptimal clinical decisions.

AIMS

To develop a system that automatically extracts, summarises, and visualises trends from longitudinal patient test reports (structured and unstructured data), and surfaces personalised best-practice clinical recommendations. Evaluated here on respiratory function tests and echocardiograms.

METHODS

A model-agnostic LLM iteratively processes patients' test reports to extract data points for key clinical metrics, plot trends and create a summary. Prompts include guardrails to avoid hallucinations. Best-practice recommendations are codified and dynamically presented via a personalised rule-based look-up.

SYSTEM OUTPUTS (PER REPORT MODALITY)

- Tabular extraction of numerical and non-numerical trends
- Synthesis: Assessment · Classification · Risk Level · Suggestions · Actions

Reports (per patient) → LLM (Analyser) → Plots & Summaries → Clinical Review

Clinical summary

Based on the pulmonary reports with known dates, the pulmonary risk appears to be increasing over time.

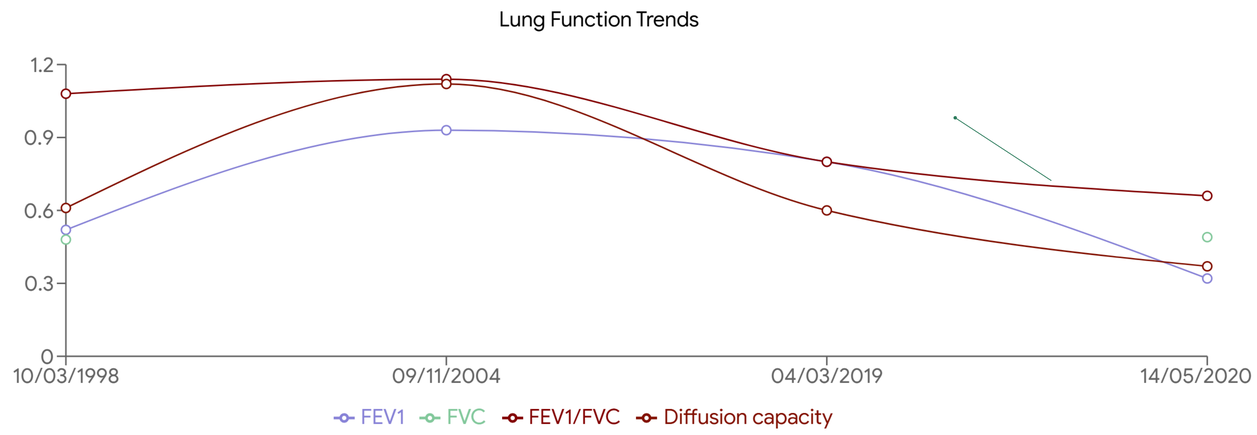
The type of pulmonary issue has been changing from restrictive to mixed, based on the reports with known dates.

Relevant risks for Pancreaticoduodenectomy/Whipples Procedure (Upper gastrointestinal surgery) that can affect anesthetic choice include pulmonary issues like restrictive lung disease, obstructive lung disease, and mixed patterns identified on pulmonary function tests. Echocardiogram findings like valvular stenosis/regurgitation, ventricular dysfunction, and pulmonary hypertension can also impact anesthetic management.

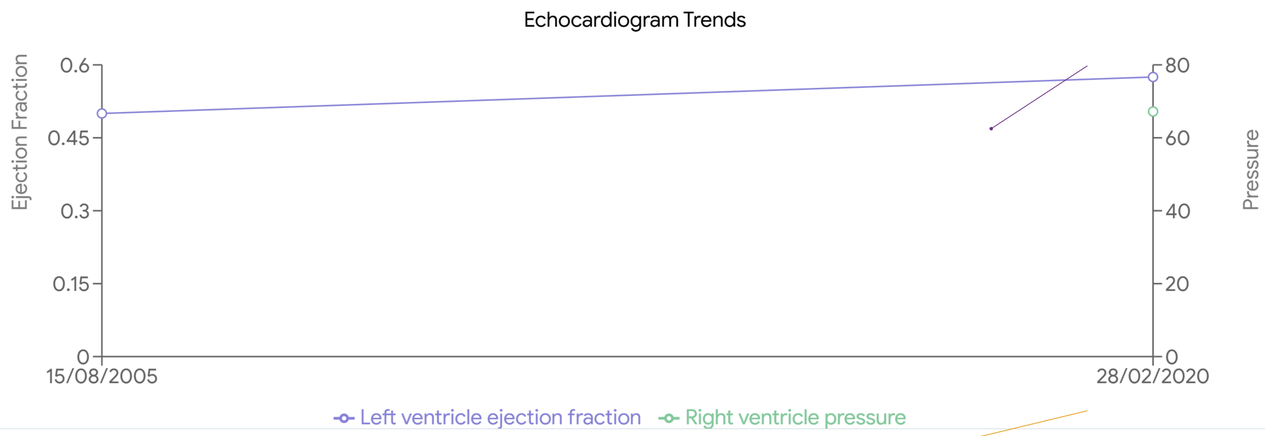
Advice based on medical knowledge

For this patient undergoing Pancreaticoduodenectomy/Whipples Procedure, I would recommend a balanced anesthetic technique with a combination of intravenous and inhalational agents, carefully titrated based on the patient's pulmonary reserve and cardiac function. Lung-protective ventilation strategies should be employed. Invasive hemodynamic monitoring may be warranted given the potential for significant fluid shifts and blood loss. Close communication with the surgical team is essential to anticipate events like clamping/unclamping and adjust anesthetic depth accordingly. Multimodal analgesia with neuraxial and regional techniques should be considered for optimal pain control while minimizing respiratory depression.

Lung function trends



Echocardiogram trends



Best-practice advice

Tabular data & synthesis

Lung Function Test Results

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Date	FEV1	FEV1/FVC	FVC	Diffusion capacity	Bronchodilators r...	Bronchodilators r...	Lung condition p...	Potential conditi...	DLCO category	Assessment	Suggestion	Lung function cla...	Risk Level	Actions
10/03/1998	0.52	1.08	0.48	0.61	minimal	0.03	restrictive	asbestosis, beryli...	low-mild	Medium risk of p...	Confirm restrictiv...		med	
14/05/2020	0.32	0.66	0.49	0.37	modest		mixed		low-severe	High risk of perio...	Lung function tes...		high	
04/03/2019	0.8	0.8		0.6	No information pr...				low-mild	Low risk of perio...	Lung function tes...		low	
09/11/2004	0.93	1.14		1.12	no significant im...				normal	Low risk of perio...	Lung function tes...		low	
unknown	1.03	0.75	1.1	1.21			normal	asthma, left-to-ri...	high	Low risk of perio...	Lung function tes...		low	

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Echocardiogram Results

Date	Left ven...	Right ve...	Left atri...	Left atri...	Right atr...	Right atr...	Left ven...	Left ven...	Right ve...	Right ve...	Pulmona...	Pulmona...	Pulmona...	Aortic v...	Aortic v...	Tricuspi...	Tricuspi...	Mitral va...	Mitral va...	Pericard...	Pericard...	Wall mot...	Wall mot...	Actions

RESULTS

The system analysed reports, extracted key information, presented trends over time in plot and table formats, provided text-based summaries and gave best-practice clinical recommendations. Results were reviewed by an experienced anaesthetist and found to be accurate.

CONCLUSIONS

This proof-of-concept demonstrates that a GenAI-based system can accurately synthesise longitudinal patient reports, visualise results, and surface personalised best-practice recommendations, significantly reducing clinician cognitive load. A prospective clinical trial is warranted.

FUTURE WORK

- RAG-based dynamic best-practice guideline lookup
- Clinician chat interface for interactive report discussion
- Prospective clinical trial evaluation